THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The area of a triangle with base 10 cm and height 8 cm is (in cm^2): 1

(a) 30 **(b)** 48 **(c)** 40 **(d)** 50

2. $\frac{1}{2} + \frac{1}{3} =$ 1

(a) $\frac{71}{78}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{5}{6}$ **(d)** $\frac{1}{6}$

3. Two circles have radii 5 cm and 3 cm and their centres are 9 cm apart. The circles: 1

(a) touch internally **(b)** one lies inside the other

(c) touch externally **(d)** do not touch (separate)

4. A rhombus (not a square) has how many lines of symmetry? 1

(a) 4 **(b)** 2 **(c)** 0 **(d)** 1

5. -12 + 7 = 1

(a) -5 **(b)** 9 **(c)** -3 **(d)** -7

6. A right-angled triangle has legs 6 cm and 8 cm. Its area (in cm^2) is: 1

(a) 32 **(b)** 18 **(c)** 30 **(d)** 24

7. $\frac{3}{4} - \frac{1}{3} =$ 1

(a) $1\frac{1}{12}$ **(b)** $\frac{5}{12}$ **(c)** 2 **(d)** $\frac{1}{3}$

8. Two points A and B are 8 cm apart. The perpendicular bisector of AB meets AB at a point which is at what distance from A (cm)? 1

(a) 4 **(b)** 3 **(c)** 8 **(d)** 5

9. The smallest angle through which a regular pentagon must be rotated to coincide with itself is (degrees): 1

(a) 62 **(b)** 82 **(c)** 72 **(d)** 102

10. 5 - 12 = 1

(a) -7 **(b)** -5 **(c)** 17 **(d)** -9

11. A square has area 64 m^2. Its perimeter (in m) is: 1

(a) 24 **(b)** 48 **(c)** 40 **(d)** 32

12. Which is greater: $\frac{3}{5}$ or $\frac{5}{8}$? 1

(a) they are equal **(b)** $\frac{5}{8}$

(c) $\frac{3}{5}$ **(d)** cannot be compared

13. The diagonals of a rectangle are each 10 cm long and meet at O. The length OA (cm) is: 1

(a) 4 **(b)** 5 **(c)** 10 **(d)** 6

14. The mirror image of the letter A in a vertical mirror to its right looks like: 1

(a) a reversed letter **(b)** an upside-down letter

(c) a different letter **(d)** the same letter

15. On the number line, the integer -6 lies 6 units to the left of 0. The integer is: 1

(a) 6 **(b)** -5 **(c)** -6 **(d)** -7

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the area of a triangle with base 12 cm and height 9 cm. 2

17. Add $\frac{1}{4}$ and $\frac{1}{6}$. 2

18. Two circles of radii 6 cm and 4 cm have centres 10 cm apart. How do they lie? 2

19. Draw (describe) all lines of symmetry of a rectangle (not square). 2

20. Show $5 + (-3)$ using tokens (positive tokens and negative tokens). Find the value. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A rectangle is 15 cm long and 8 cm wide. Find its perimeter and area. 3

22. Find the sum of $\frac{2}{3}$ and $\frac{3}{4}$ and compare the sum with 1. 3

23. Describe the construction of the perpendicular bisector of a line segment of length 6 cm. 3

24. A regular polygon has rotational symmetry of angle 40°. How many sides does it have, and how many lines of symmetry? 3

OR

Evaluate $8 + (-12) + 5$. 3

25. Find the area of a right-angled triangle with sides 6 cm, 8 cm and hypotenuse 10 cm, and its perimeter. 3

OR

3 metres of cloth is cut into pieces of $\frac{2}{5}$ m each. How many pieces are made and how much is left? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. An L-shaped figure is made by joining a rectangle 10 cm by 6 cm and a rectangle 4 cm by 5 cm. Find the total area and the area if the smaller part were removed. 5

OR

Riya reads $\frac{1}{4}$ of a book on Monday and $\frac{1}{3}$ of it on Tuesday. What fraction has she read in two days? What fraction is left? On which day did she read more? 5

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27. Construct a rectangle with sides 8 cm and 6 cm using only a ruler and compass (describe the steps). Verify by measuring that the diagonals are equal (state the property used). 5

OR

Study the shape: a regular hexagon. Write 5

(a) number of lines of symmetry,

(b) order of rotational symmetry,

(c) smallest angle of rotation,

(d) the angle after 4 such rotations.

END OF QUESTION PAPER **Total: 50 Marks**

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